The Semantic Fit of Modifiers Influences Noun Retrieval During Reference Production

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Introduction: Compared to unmodified noun phrases (e.g., *the king*), modified noun phrases (e.g., *the cruel king*) tend to be referred to using more attenuated referring expressions such as pronouns [7, 8, 9]. This suggests modifications facilitate memory retrieval of the associated noun during language production [1, 2, 3]. Under memory-based accounts of language processing [10, 12], modifiers aid retrieval because they create distinct memory representations of nouns by providing more informative retrieval cues [5, 6, 11, 13]. However, semantically unfit modifiers (*kind and merciful dictator*) tend to result in a retrieval *cost,* contrary to what cue-based retrieval theories would predict [5]. The current study aimed to investigate if the semantic fit between modifiers (*holding a crown* vs. *holding a potion*) and two nouns in a sentence (*king* vs. *witch*) could impact subsequent retrieval during language production, using experimental sentences such as those in Table 1. If "modifier fit" impacts the distinctiveness of the resulting noun representation, greater semantic fit between a noun and the associated modifiers should lead to greater rates of pronominal reference to that noun, suggesting easier retrieval from memory. If modifier fit does not affect the distinctiveness of associated nouns, pronominal reference rates should not vary by semantic fit. **Method**: We ran a norming study to quantify how well the modifiers semantically fit each noun. Forty-five participants used a continuous sliding scale (1 – 5) to judge how well the modifiers semantically fit either noun, where smaller numbers on the scale indicated greater semantic fit between the modifiers and the first-mentioned noun (NP1), and greater numbers reflected greater semantic fit between the modifiers and the second-mentioned noun (NP2). Numbers in the middle of the scale (around 3) indicated nearly equal fit. For the main study, 137 participants were presented with single sentences and asked to type a single sentence that would "continue the story". There were 60 experimental sentences and 80 fillers. Experimental sentences either modified NP1 or NP2 (see Table 1), and modifiers had varying degrees of semantic fit to the associated noun, as assessed by our norming experiment. We coded both *which* noun was referred to (i.e., *NP Choice*: NP1 vs. NP2) and *how* the chosen noun was referred to (i.e., *Referential Form*: repeated-name reference (*king)*, vs. pronominal reference (*he)*) using the guidelines presented in Table 2. **Results**: Using a logit mixed-effects regression model, we analyzed if Referential Form and/or NP Choice was predicted by the continuous variable of Modifier Fit (1 – 5). Models included random slopes for participants and items, as well as by-subjects and by-items random slopes for the effect of Modifier Fit, as well as its interaction with NP Choice. Consistent with prior findings, NP1 was significantly more likely to be pronominalized than NP2 [4]. Critically, there was a significant interaction between NP Choice and Modifier Fit (see Table 3 and Figure 1). For NP1, increasing Modifier Fit values (i.e., greater semantic fit with NP2) resulted in smaller rates of pronominal reference (left panel of Figure 1). For NP2, increasing Modifier Fit values (i.e., greater semantic fit with NP2) resulted in greater rates of pronominal reference. **Discussion**: The results suggest that the semantic content of modifiers affects subsequent retrieval difficulty for the associated nouns. When modifiers semantically fit a noun, pronominal references to that noun increased, indicating a modification-induced retrieval benefit. In other words, as modifiers grew to semantically fit the competitor, the retrieval benefit to the associated noun was lost. This has important implications for theories of distinctiveness and cue-based retrieval [11, 12], as retrieval seems to be influenced not only by cues intrinsic to the nouns themselves [+male], but also by extrinsic cues that modifiers can provide [+holding a crown].

Table 1. Stimuli Examples

The king who was cruel and merciless criticized the witch.
The king criticized the witch who was cruel and merciless.
The king who was holding the crown criticized the witch.
The king criticized the witch who was holding the potion.

Table 2. Data Coding Procedure

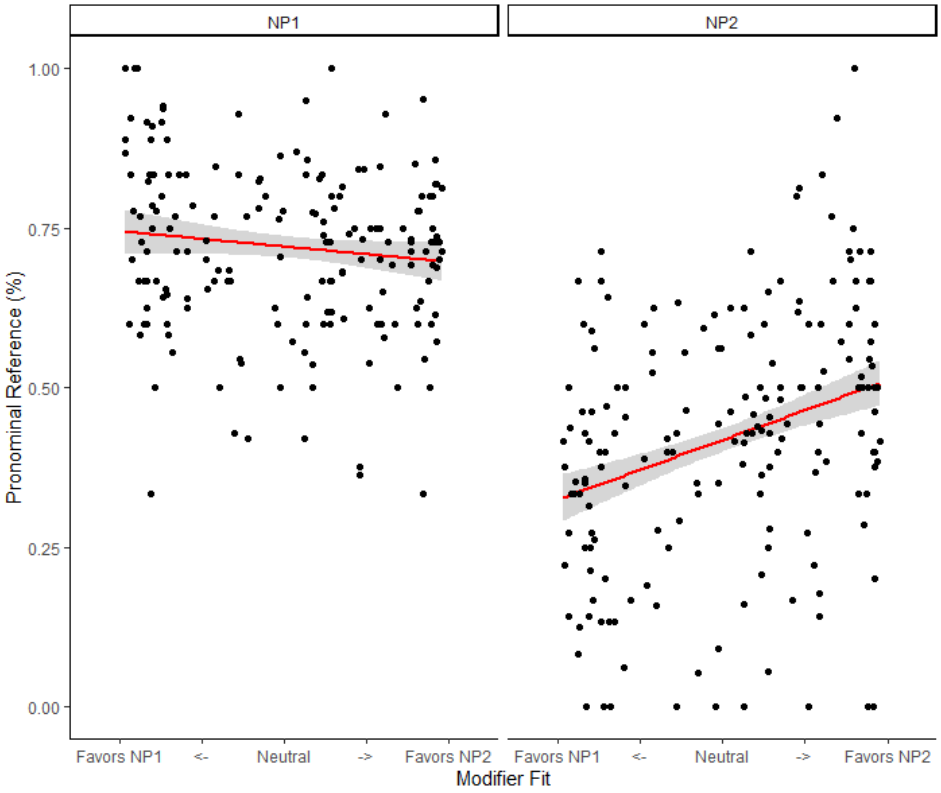
Response Type	Response Sentence
Pronominal Reference (1)	
NP Choice: NP1	*He* was quite annoyed.
NP Choice: NP2	*She* was sick of it all.
Repeated Noun Reference (0)	
NP Choice: NP1	*The king* was full of hate.
NP Choice: NP2	*The witch* was ready to retire.
Collective Reference (NA)	
Named	*The king and the witch* were sad.
Pronoun	*They* had a vicious spat.
Invalid Responses (NA)	
Possessive Adjective	*His sanity* was at its limit.
Not First Referent	*The kingdom* he had was terrified.
Subordinate Clause	*When he left, the witch* laughed.
New Noun Referent	*Dissent* is unacceptable.

Table 3. Effects of predictors on pronominal vs. repeated noun referent production

Predictor	Estimate (SE)	*Z*	*p*
Choice of NP (NP1 vs NP2)	-2.442 (0.184)	-13.215	<0.001
Modifier Fit (1 – 5)	0.089 (0.057)	1.560	0.119
Interaction	0.611 (0.121)	5.062	<0.001
Modifier Fit effect for NP1	-0.274 (0.106)	-2.587	0.009
Modifier Fit effect for NP2	0.389 (0.078)	4.997	<0.001

Note: NP1 is coded as -0.5, NP2 is coded as 0.5

Figure 1. Proportion of Pronominal References to Each Noun Phrase



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